

# Kyuhwan Shim

Graduate School of Artificial Intelligence  
Seoul National University  
Haedong Advanced Engineering Bldg. #303  
Gwanak-ro 1, Gwanak-gu, Seoul, Republic of Korea 08826

✉ [kyuhwan.shim@snu.ac.kr](mailto:kyuhwan.shim@snu.ac.kr)  
✉ [kyuhwan.shim@lgresearch.ai](mailto:kyuhwan.shim@lgresearch.ai)  
🌐 <https://underthelights.github.io>  
in [kyuhwan-shim](#)

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|---------------------|----------------------------------------------------------------------------------------------------|---------------------|
| EDUCATION           | <b>Seoul National University</b> , Seoul, Korea                                                    |                     |
|                     | Master Student in Graduate School of Artificial Intelligence                                       | Sep 2024 – Present  |
|                     | <b>Sogang University</b> , Seoul, Korea                                                            |                     |
|                     | Bachelor of Engineering in Computer Science                                                        | Mar 2019 – Aug 2024 |
| RESEARCH EXPERIENCE | <b>Chungnam Science High School</b> , Chungnam, Korea                                              | Mar 2016 – Feb 2019 |
|                     | <b>LG AI Research</b>                                                                              |                     |
|                     | Research Scientist Intern, Physical Intelligence Lab                                               | Nov 2025 – Present  |
|                     | <b>Universität Bonn, Deutschland</b>                                                               |                     |
| WORK EXPERIENCE     | Visiting Student Researcher, Autonomous Intelligent Systems Group                                  | Jul 2024 – Aug 2024 |
|                     | <b>Seoul National University</b>                                                                   |                     |
|                     | Undergraduate Student Researcher, Biointelligence Lab                                              | Dec 2023 – Aug 2024 |
|                     | ExploreCSR, Graduate School of Data Science & Google Research                                      | Dec 2023 – Feb 2024 |
| WORK EXPERIENCE     | <b>TIDYBOY-DSPL</b> , Seoul, Korea                                                                 | Mar 2024 – Aug 2024 |
|                     | Vision Engineer                                                                                    |                     |
|                     | ▪ RoboCup 2024 team from Seoul National University and Pusan National University                   |                     |
|                     | ▪ Developed a robot vision pipeline for real-time environment understanding                        |                     |
| WORK EXPERIENCE     | <b>Nota Inc.</b> , Seoul, Korea                                                                    | Mar 2023 – Aug 2023 |
|                     | Machine Learning Engineer Intern                                                                   |                     |
|                     | ▪ Developed vision-based pipeline models for intelligent traffic systems on edge devices           |                     |
|                     | ▪ Constructed data lake and ML pipelines on AWS and GCP                                            |                     |
| WORK EXPERIENCE     | <b>OCONNECT</b> , Ulsan, Korea                                                                     | Jun 2020 – Feb 2023 |
|                     | Co-Founder, Chief Design & Operating Officer                                                       |                     |
|                     | ▪ <b>[LINK]</b> UNIST-based student venture developing ONE STEP, an easy-insert socket             |                     |
|                     | ▪ R&D Didimdol, James Dyson Award, Pre-Startup Package, Youth Entrepreneurship Academy             |                     |
| WORK EXPERIENCE     | <b>Yeongdong Firestation</b> , Yeongdong, Korea                                                    |                     |
|                     | Conscripted Firefighters (Military Duty)                                                           | Aug 2020 – Apr 2022 |
| AWARDS & ACTIVITIES | <b>Encouragement Award</b> , Seoul National University                                             | 2024, 2025          |
|                     | <b>Finalist (Top 3)</b> , RoboCup Federation                                                       | Aug 2024            |
|                     | <b>Finalist (Top 2)</b> , Seoul National University                                                | 2024                |
|                     | <b>Top 30 &amp; Talent Pool</b> , Samsung Advanced Institute of Technology & DAEON                 | Sep 2023            |
| AWARDS & ACTIVITIES | <b>Top 30</b> , LG AI Research & Ministry of Employment and Labor                                  | Sep 2023            |
|                     | <b>Grand Prize</b> , Gangseo-gu Office                                                             | 2023                |
|                     | <b>Grand Prize</b> , National Information Society Agency, Ministry of Science and ICT              | 2022                |
|                     | <b>Excellence Award</b> , Chungnam Science Fair (64th)                                             | 2018                |
| AWARDS & ACTIVITIES | <b>Special Prize</b> , Chungnam Science Fair (63rd)                                                | 2017                |
| SCHOLARSHIPS        | <b>SNU AI Fellowship</b> , Seoul National University                                               | 2024                |
|                     | <b>Sogang University Alumni Scholarship</b> , Sogang University Alumni Association                 | 2022-2024           |
|                     | Priest Villarreal Memorial Smilegate Scholarship, General Scholarship, Startup Talents Scholarship |                     |
|                     | <b>Albatross SW Scholarship</b> , Sogang University                                                | 2019                |
| MEDIA               | <b>Excellence Paper Award in the Undergraduate Division at KCC 2024</b> , Sogang People            | Jun 2024            |
|                     | <b>“Nunettine” wins Grand Prize at Gangseo-gu Big Data Contest</b> , Sogang People                 | May 2023            |
|                     | <b>“DAVENGERS” wins BIGCONTEST Data Analytics Innovation League</b> , Sogang People                | Dec 2022            |

## PUBLICATIONS

## INTERNATIONAL CONFERENCE

- [C4] **Beyond Fixed Tasks: Scaling Robot Data with Region-Aware Interaction Network**, Seunghun Lee\*, **Kyuhwan Shim\***, Yeogyong Song\*, Seung-Hwan Kim, Sihaeng Lee, Byoung-Tak Zhang<sup>†</sup>, Sunghoon Lim<sup>†</sup>, Gunhee Lee<sup>†</sup>, *Conference of Robot Learning (CoRL) 2026 (Under Review)*
- [C3] **Making Your Action Policies Interpretable: Mixture of Action Queries**, Suhjung Choi, Youngseok Joo, Hyundo Lee, Kisung Shin, **Kyuhwan Shim**, Chungwoo Lee, Minjeong Gu, Jun Ki Lee, Byoung-Tak Zhang<sup>†</sup>, *Foundation Models Meet Embodied Agents @ CVPRW 2026, AI4GOOD @ ICMLW 2026*
- [C2] **ESPADA: Execution Speedup via Semantics Aware Demonstration Data Downsampling for Imitation Learning**, Byungju Kim, Pahk Jinu, ChungWoo Lee, Jaejoon Kim, Jangha Lee, Theo Taeyeon Kim, **Kyuhwan Shim**, Jun Ki Lee, Byoung-Tak Zhang<sup>†</sup>, *IEEE Robotics and Automation Letters (RA-L) 2026*
- [C1] **CLIP-RT : Learning Language-Conditioned Robotic Policies from Natural Language Supervision**, Gi-Cheon Kang\*, Junghyun Kim\*, **Kyuhwan Shim**, Jun Ki Lee<sup>†</sup>, Byoung-Tak Zhang<sup>†</sup> *Robotics: Science and Systems (RSS), 2025 (Oral)*

## DOMESTIC JOURNALS

- [D4] **Scaling Language-Conditioned Policies with Fast and Lightweight Architectures**, Junghyun Kim\*, Gi-Cheon Kang\*, **Kyuhwan Shim**, Eunseo Kim, Jun Ki Lee<sup>†</sup>, Byoung-Tak Zhang<sup>†</sup> *Korea Computer Congress 2024*  
🏆 Excellent Presentation Award
- [D3] **Generalizing Language-Conditioned Robotic Policies via Large Language Models**, Junghyun Kim\*, Gi-Cheon Kang\*, **Kyuhwan Shim**, Eunseo Kim, Jun Ki Lee<sup>†</sup>, Byoung-Tak Zhang<sup>†</sup> *Korea Computer Congress 2024*  
🏆 Excellent Presentation Award
- [D2] **Stochastic Trajectory Diversification for Language-based Robot Action Learning**, Junghyun Kim\*, Gi-Cheon Kang\*, **Kyuhwan Shim**, Jun Ki Lee<sup>†</sup>, Byoung-Tak Zhang<sup>†</sup> *Korea Software Congress 2024*  
🏆 Excellent Presentation Award
- [D1] **KIRINO: An Interactive Chatbot System for User Persona**, Geon Choi\*, Ganghun Kim\*, Hyunjae Kim\*, **Kyuhwan Shim\***, Myoungwan Koo<sup>†</sup> *Korea Computer Congress 2024*  
🏆 Excellent Paper Award

TEACHING  
EXPERIENCE

Artificial Intelligence (4190.408), Seoul National University

Mar 2025 – Jun 2025

Translation Volunteer, Khan Academy &amp; Chungnam Science High School

Jun 2017 – Dec 2017

## SPEECH

- [PDF] Seminar Samsung DICON Challenge - Domain Adaptation @ Release(Sogang CS) Dec 2023
- [PDF] Seminar Dive into YOLOv8 @ Nota Inc. Mar 2023
- [PDF] Undergraduate Speech at Carbon Neutral Workshop @ Shinhan Card & World Bank 2023

## ENTREPRENEUR

**OCONNECT** - a student venture based at UNIST, founded to address usability challenges in everyday life. The team developed **ONE STEP**, a rotating, easy-to-plug-in electrical outlet designed to enable more intuitive and convenient plug insertion.

## Awards

- **James Dyson Award 2022 - Nat'l, Korea**, James Dyson Foundation
  - Winner (Top 3), International Competition Jul 2022
- **PIN UP Concept Design Award 2021**, KAID
  - BEST 100 in Living Goods Design Dec 2021
- **Busan-Ulsan-KyungPook Startup Day Innovation Ventures League**
  - Samsung Electronics; KyungPook, Ulsan, Busan Creative Econ. & Innov. Center
  - Excellence Prize at Futures' League Dec 2021

- **Challenge K-Startup Expo : Collegiate Competition**  
Ministry of SMEs and Startups,  
Korean Institute of Startup & Entrepreneurship Development
  - Top 10, Offline Exhibition Oct 2021
  - Top 50, Offline Exhibition Aug 2019
- **Social Challenge alpha 7**, OhMyCompany, BSS, LISCC
  - Challenge Award, Crowd Funding (Funded 1,424,000 KRW) Sep 2020 – Dec 2020
- **Ulsan Collegiate Startup Excellent Idea Contest**  
Ulsan Creative Econ. & Innov. Center
  - Winner: Grand Prize Nov 2020
- **Institute of Science & Technology Startups Competition**  
KAIST, GIST, DGIST, UNIST, POSTECH
  - Representative of UNIST (Top 2 at UNIST) Dec 2020
  - Representative of UNIST (Top 3 at UNIST) Nov 2019

## Funding & Support Programs

- **DIDIMDOL First Step - Startup Growth Development Project**,  
Ministry of SMEs and Startups,
  - Recommended from Ulsan Creative Econ. & Innov. Center Aug 2022 – Jul 2023
- **Ulju Youth Startup Academy**, Ulsan Youth Startup Center Mar 2022 – Dec 2022
- **Pre-IBK Changgong @ UNIST**, UNIST, IBK Changgong Jul 2022 – Nov 2022
- **Private-Sector Open Start-up Revitalization Project**,  
Ulsan Creative Econ. & Innov. Center May 2022 – Oct 2022
- **BUSAN BIGS Program**,  
Busan-si, KSD, KHUG, KOSPO, HF, KIBO, Busan PA, BNK Mar 2022 – Sep 2022
- **Innopolis Campus for Ulsan Collegiate**, UNIST, Innopolis Jun 2021 – Aug 2022
- **12<sup>th</sup> UEPA Startup Incubator**, Ulsan Youth Startup Center Mar 2021 – Dec 2021
- **SBC Youth Entrepreneurship Academy**, Korea SMEs and Startups Agency
  - Most Funded Startup Company (100,000,000 KRW)
  - Excellent Completion at Pre-School
  - Pre-School Completion of Excellence & Youth Startup Academy Nov 2020 – Nov 2021
- **UNIST UNicorn Start-Up Support Program**, UNIST
  - Funding (12,000,000 KRW)
  - Participated for four semesters Sep 2019 – Feb 2021
- **KISED Pre-Startup Package (General Field)**,  
Korea Institute of Startup & Entrepreneurship Development
  - Most Funded (70,000,000 Won) Apr 2020 – Dec 2020

## Media Coverage

- **UNIST Startup OCONNECT wins at JAMES DYSON AWARD 2022**,  
UNIST Press, CHOSUN Ilbo, Segye Ilbo, Sisa-news, Newsis Feb 2022
- **UNIST 2021 Innopolis Performance Presentation**,  
UNIST Press, UlsanDaily, ETNews Feb 2022
- **UNIST UNISPARK for Young Entrepreneurs** UNIST Magazine 2021 Winter Dec 2021
- **Southeast Venture Competition held by UNIST Lab Startup Innovation Group**  
The Korea Economy TV Dec 2021
- **"New evolution of electrical outlets..." OneStep is released**  
The Korea Economic Daily Dec 2021
- **Make a National Outlet which Connects warmth to each other**  
UNIST Magazine 2021 Spring Feb 2021
- **Doing businesses regardless of COVID-19 - Leap forward with technology**  
SBS, UBC Feb 2021